

Giovani Diaz Alfaro

Cognitive Scientist

Orlando, FL – Open to Relocation

Creative and detail-oriented scientist with 6 years of experience in interdisciplinary research and development. Experienced in project management, workflow development, research study execution, data analysis, and publication with a strong focus on synthesizing knowledge from multiple disciplines to address complex, challenging issues and questions.

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EDUCATION

**MSc Interdisciplinary Studies
(conc. Cognitive Systems, Ethics)**
University of Central Florida
Coursework Complete —
Conferral Pending, 2026

GradCert. Cognitive Sciences
University of Central Florida
Spring 2023

**BA Philosophy
(min. Cognitive Sciences)**
University of Central Florida
Spring 2020

SOFTWARE

Scripting and Statistics: Python, R,
MATLAB

Graphics and Diagrams: Adobe
Photoshop

Webdev: JavaScript, WordPress,
basic HTML5

Surveys and Studies: Qualtrics

Publishing: LaTeX

LANGUAGES

Bilingual: English • Spanish

Basic Proficiency: French

PROFESSIONAL EXPERIENCE

Researcher
Cognitive Sciences Laboratory, University of Central Florida | Orlando (FL), USA
June 2023 – Present

United States Department of Defense, NASA, Minerva Research Initiative:
Understanding Cognition in Support of Future Space Missions
(PI: S.M. Fiore)

- Develop unobtrusive measures of team functioning for future space missions by leveraging multi-sensor data (e.g., physiological, environmental, etc.) to monitor individual and team cognitive states.
- Implemented suggestions to enhance operational environment workflow and design for long-duration space exploration missions with human and artificially intelligent (e.g. onboard systems) team members.

Lockheed Martin Corporation: Externalized and Extended Cognition: *Cognitive Offloading for Manned-Unmanned Teaming*
(PI: S.M. Fiore)

- Provide recommendations for mitigating workload in human-machine teaming contexts (e.g. AI as teammate), as well as offer a multi-disciplinarily informed set of theoretically grounded research questions and designs for understanding and improving workload in manned-unmanned teaming environments.

- Conduct interdisciplinary research on individuals and teams to evaluate cognition and develop interventions to reduce workload and improve performance.
- Interpret and understand the specific needs and problem areas for industries /companies such as the United States Department of Defense, Lockheed Martin, the United States Air Force, and more.
- Perform research focused on extended cognition, its theoretic foundation and principles, and practical applications in contexts such as virtual reality, space travel, military operations, and more.
- Apply to and analyze the role of ethical theories and principles in complex contemporary situations.
- Perform and apply human-machine systems research principles in interdisciplinary contexts.
- Manage lab teams and experiment logistics needs, human subjects for research, and maintenance of lab/experimental environments.
- Propose research for funding, design and conduct experiments, and analyze complex data with the end goal to publish results.
- Train in, and prepare for future work in teaching, research, project management, and consulting.

KEY SKILLS

Experimentation:

Human-Subjects Research, Design, Planning, Workflow Management, Surveying, Quality Assurance Testing

Collaboration:

Multi-disciplinary research, multi-disciplinary project management

Soft Skills:

Public Speaking, Collaboration, Creative Thinking, Concept Presentation

Other:

Team Member/Employee Management, Computer Hardware

CERTIFICATIONS

Python 3 (Codecademy, 2022)

Inclusive and Ethical Leadership Certificate (Leadership Summit & USF, 2024)

RESEARCH INTERESTS

- Practical Cognitive Science
- Human Enhancement
- Team Cognition
- Interdisciplinarity
- Human-Machine Teaming
- Software for Scientific Teams
- Moral Cognition
- Ethics of Artificial Intelligence
- Critical Pedagogy

Supervisor, Process Improvement Coordinator

Orlando Health Imaging Centers | Orlando (FL), USA

Sep 2020 – Dec 2021

Project(s): Quality Assurance Program, St. Petersburg Expansion, EPIC Systems Launch and Training

- Conducted task analysis across clinical and non-clinical workflows to identify failure points and inefficiencies; defined performance metrics and led organization-wide process-redesign initiatives to reduce error and improve safety.
- Analyzed operational data against internal and external benchmarks to identify human-error risk points in clinical workflows; designed and implemented process interventions that reduced errors by 60% and increased departmental productivity by 40%.
- Led cross-functional teams through training and communication structures supporting safe, reliable execution of redesigned clinical workflows.
- Monitored operational data trends to surface workflow optimization opportunities; reported productivity, error, and revenue metrics across departments using Excel to support evidence-based process decisions.
- Conducted ongoing process audits — gathering data, developing workflow diagrams, and presenting findings to stakeholders — to build buy-in for human-factors-informed process changes.

Reviewer

HFES International Annual Meeting

2024– 2026 (present)

PROFESSIONAL DEVELOPMENT

Team Learning for Better Scientific Software (TL4BSSw)

Brought together experts across disciplines and domains to develop an open knowledge base with tools to support and evaluate team learning in the interdisciplinary groups who produce scientific software. The products of the workshop contribute to understanding of collaborative learning in scientific software development and use, and are open to all in service of better scientific software.

Related manuscript in preparation:

Newton, O. B., **Diaz Alfaro, G.**, Raybourn, E. M., Curfman McInnes, L., Lenhardt, C., Aydeniz, M., Frluckaj, H., Zhou, H., Milewicz, R., Serhan, Z., Almutairi, M., Ramchandra Patole, S. (in preparation). Team learning for better scientific software.

Target: IEEE Computing in Science and Engineering

EPIC System Launch (Orlando Health Imaging Centers)

Managed, trained, and led resource construction for the organization-wide launch and integration of EPIC's healthcare software system.

RELATED EXPERIENCE

Guest Lecturer

Topics in Cognitive Science (Fall '24/'25)

[UCF, Instructor: S. M. Fiore; ≤15 graduate students]

Research Methods in Cognitive Science (Spring '25)

[UCF, Instructor: S. M. Fiore; ≤15 graduate students]

Teaching Assistant

Ethics in Science & Technology (Spring '20)
(UCF, Instructor: D. Jones)

Mentorship

Sophia Sakakibara Capello, *B.Sc. Psychology, Minor: Cognitive Sciences* / Accepted to STEM Summer Scholars Institute at Indiana University Bloomington (IUB) & Indiana University-Purdue University Indianapolis (IUPUI)

Reviewer

Conference: Human Factors and Ergonomics Society International Annual Meeting (2024-26)

PUBLICATIONS

Conference Proceedings

Diaz Alfaró, G., Fiore, S. M., & Oden, K. (2024). Externalized and extended cognition: Cognitive Offloading for human-machine teaming. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*.

<https://doi.org/10.1177/10711813241275506>

Presentations, Panels, and Posters

Diaz Alfaró, G. (2024, November). Team Learning for a Better Foundation for Scientific Software. Position Paper presented at the 2024 Team Learning for Better Scientific Software Workshop, Orlando, FL.

Sonnenfeld, N., Nguyen, B., **Diaz Alfaró, G.**, Daniliv, V., Newton, O. B., Jentsch, F., Fiore, S. M. (2024). Eliciting Requirements and Recommendations for Training Macrocognition in Teams: Considerations for Collaborative Problem Solving. *Poster presented at 2024 ASPIRE—the HFES International Annual Meeting, Phoenix, AZ.*